API Protocol: Auth

Duotecno	Showroom:	T +3250 200560
Astridlaan 170A	Ten Briele 18.1	F +3250 200561
8310 Assebroek	8200 Sint-Michiels	
Belgium	Belgium	email: info@duotecno.be

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API

AUTHORIZATION

Version 1.05

Info: http://www.duotecno.be

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Overview Authorization API

The base url for this API is: **https://[IP-Address]:8081/**

Protocol: https.
[IP-Address]: The IP-Address of the Duotecno node.
The port is always 8081.

Possibilities:
- Create client.
- Refresh client token.

The number of client login credentials is limited.
A client must only create the first time his login credentials. He must store these credentials and reuse them. When the access token is expired he must ask a new token via the refresh request.

To reduce the number of clients in the server a client MUST first always try to refresh his access token. **Only when this is not possible he must create a new client id.** This can happen when the client is not used for a long time.
When the server holds many clients an needs space for newer client ids it will remove the oldest clients first.

The client id and access tokens should be used in the configuration API.
A client id will only be stored by the server when it uses the configuration API.

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.05.08
TCP/IP Homeserver	V66.05.08
Universal audio interface	V66.05.08

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Create client credentials

First a client must be created.
The response holds the client id and access tokens to use in the HTTP header of the configuration API functions.

Once a client is created he must use the access & refresh tokens.

Overview:

http method	GET
Endpoint	/auth/create
Response	Status code 200 (or error 400/401)

Request:

The HTTP header must hold Base64 client & password values.
The client id should be a dummy value.

Syntax: Base64 { dummyClientId:password }
Client id and password must be separated by a ':'
A ':' cannot be used in the client & password.

Response:

When successful the HTTP status code will be 200 and the json body holds the credentials to use.
When an error occurs the response will be HTTP status code 400/401. (See authorization errors).

keys	**Value**
"access_token"	String: The access token. This value should be used in the http header of the configuration API. Syntax Base64 { clientid:accesstoken } This value changes when a new access token is requested.
"client_id"	String: The client id. This value should be used in the http header of the configuration API. Syntax Base64 { clientid:accesstoken } This value never changes and must be used in the header with every call.
"expiry_time"	Integer value: The time in seconds the accesstoken is valid. After this time the client can ask a new accesstoken using the refresh_token.
"refresh_token"	String: The refresh token. This value should be used in the auth/refresh request. Syntax Base64 { clientid:refreshToken } This value changes when a new access token is requested.
"token_type	String: value is always "Bearer".

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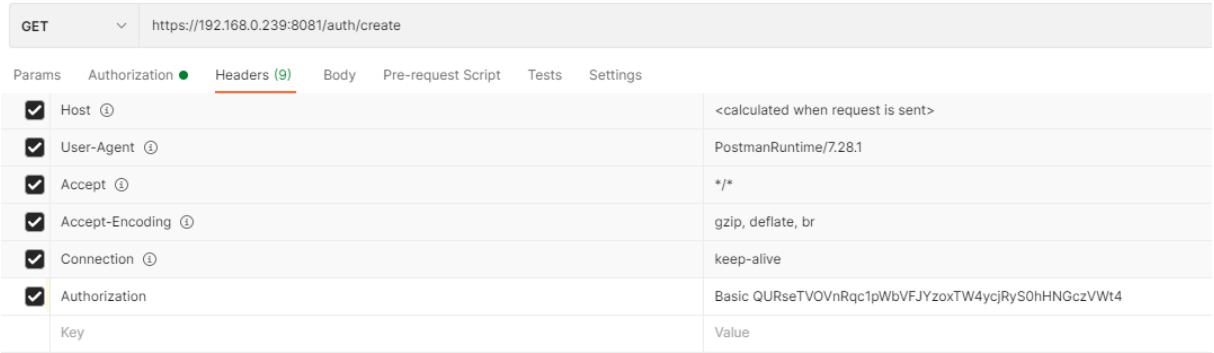
Example postman:

Request:

https://192.168.0.239:8081/auth/create
dummy client: ADly5NVtjsZVmQIc
paswoord: duotecno

Base 64 Authorization string is: QURseTVOVnRqc1pWbVFJYzpkdW90ZWNubw==



Response:


```
{
        "access_token": "xBErAnB6C3Fqu41I",
        "client_id": "aR8NtDjryC3jjWOY",
        "expiry_time": 2400,
        "refresh_token": "9lvnH7xFIlwzN8AE",
        "token_type": "Bearer"
}
```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Refresh client access token

After the expiry time the client access token can no longer be used. The client must call this request function to get a new access token.
The response holds the client id and new access token.
After this call the old access and refresh tokens can no longer be used.

In the configuration API:
When the client receives a http resonse error status code 401 “expired_token” it must use this request function to refresh his token. After refreshing the token he must use the new access token in the configuration API calls.

Overview:

http method	GET
Endpoint	/auth/refresh
Response	Status code 200 (or error 400/401)

Request:

The HTTP header must hold Base64 client & refresh access token.

Syntax: Base64 { clientid:refreshToken }
Client id and the refresh token must be separated by a ‘:’

Response:

When successful the HTTP status code will be 200 and the json body holds the new access and refresh tokens.
When an error occurs the response will be HTTP status code 400/401. (See authorization errors).

The same parameters are used as in the create reponse.
Both the “access_token” and “refresh_token” are new values.

Example postman:

Request:

https://192.168.0.239:8081/auth/refresh
client: aR8NtDjryC3jjWOY
refresh token: 9lvnH7xFIlwzN8AE

Base 64 Authorization string is: YVI4TnREanJ5QzNqaldPWTo5bHZuSDd4Rklsd3pOOEFF

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GET https://192.168.0.239:8081/auth/refresh

Params Authorization Headers (8) Body Pre-request Script Tests Settings

☑	Host	<calculated when request is sent>
☑	User-Agent	PostmanRuntime/7.28.1
☑	Accept	*/*
☑	Accept-Encoding	gzip, deflate, br
☑	Connection	keep-alive
☑	Authorization	Basic YVI4TnREanJ5QzNqaldPWTo5bHZuSDd4Rklsd3pOOEFF
	Key	Value

Response:


```
{
        "access_token": "byiD9NrK95oELNjz",
        "client_id": "aR8NtDjryC3jjWOY",
        "expiry_time": 2400,
        "refresh_token": "jVn1K1zwEBJvCkPc",
        "token_type": "Bearer"
}
```


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Authorization error codes:

These error responses are send in the authentication & configuration API.

Status code	Message	Description
400	max_clients_reached	Maximum number of clients reached.
401	invalid_request	Authentication header missing.
401	unknown_token	Unknown access of refresh token.
401	expired_token	The token is expired. Use the refresh token to create a new access token.
401	unknown_client	Unknown client
401	unknown_password	Unknown password. Response in the /auth/create request.
401	not_implemented	When a request is unknown in the https api.
401	general_error	Possible error in the server. Check the logfile of the node for more information.

The error response message holds no content in the body.

How to use in the configuration API.

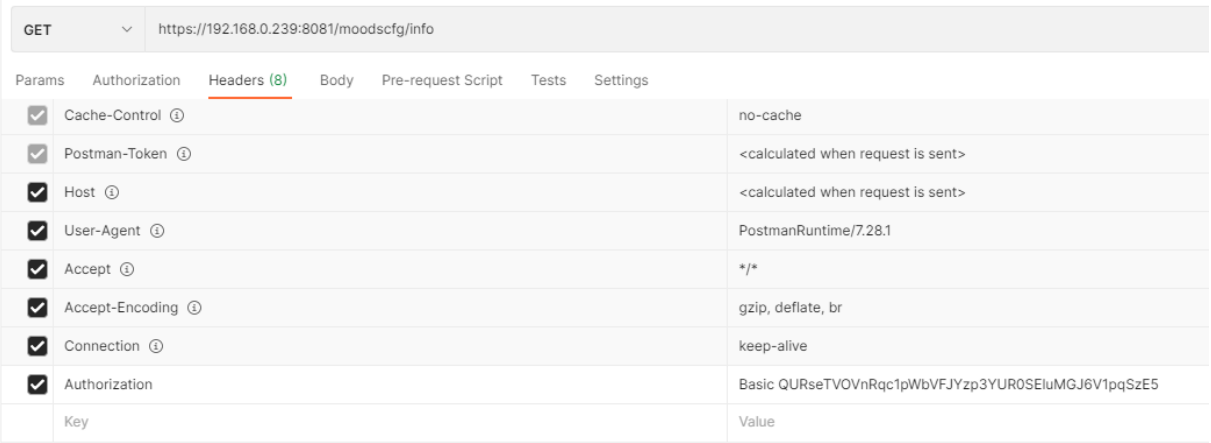
In the configuration API the http header must hold the client id and the access-token.
When the header is missing or holds wrong credentials an error reponse code will be returned.

Best practice is when the client application starts to request a new access token and refresh this token before it expires.

Example Postman: API configuration request.

https://192.168.0.239:8081/moodscfg/info
Client id: ADly5NVtjsZVmQIc
Access-token: waDtHIn0bzWZjK19

Base64 authorization string: QURseTVOVnRqc1pWbVFJYzp3YUR0SEluMGJ6V1pqSzE5



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